

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

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FINAL EXPLANATION: BIOLOGY GRADE 10 — SUB-STRAND 2.3: PLANT GASEOUS EXCHANGE AND RESPIRATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION ASSESSMENT — Student Instructions

Your task is to write a comprehensive Final Explanation that answers the unit's Driving Question:

"How do plants 'breathe,' and what happens inside their cells when they release the energy stored in food?"

Think back to the very first lesson: Mwalimu Njoroge's class submerged freshly cut sukuma wiki leaves in warm water and watched tiny bubbles stream from specific points on the leaf surface. Every section below should connect back to that observation and build toward a complete scientific explanation.

HOW TO USE THIS DOCUMENT:

- Read each section prompt carefully before you write.
- Use the exemplar response as a model — notice how it cites evidence, uses precise vocabulary, and links back to the phenomenon.
- Your own answer must be written in your own words. Do not copy the exemplar.
- Use diagrams where prompted. Label all structures clearly.

- Aim for depth over length — one well-explained idea is better than three vague ones.
- Review the rubric at the end so you know exactly what "Excellent" looks like before you start writing.

Section 1: The Sites of Gaseous Exchange — Where Do the Bubbles Come From?

Explain where gases enter and leave a plant leaf. In your answer:

- Identify and describe the specific structures responsible for gaseous exchange in leaves, stems, and roots.
- Explain why bubbles in Mwalimu Njoroge's experiment appeared from specific spots rather than from the entire leaf surface.
- Draw and label a cross-section of a leaf showing all relevant gaseous exchange structures.
- Describe how the structure of each site is suited to its function.

When the sukuma wiki leaves were submerged in warm water, the bubbles did not stream from random spots — they emerged from tiny pores called stomata (singular: stoma). Stomata are microscopic openings found mainly on the lower epidermis of leaves. Each stoma is flanked by two specialised guard cells, which are kidney-shaped and contain chloroplasts. The guard cells control the size of the stomatal opening: when they absorb water and become turgid, they bow outward and the pore opens; when they lose water and become flaccid, the pore closes. This opening and closing regulates how much CO₂ enters and how much O₂ and water vapour exit the leaf.

Inside the leaf, gases diffuse through large air spaces in the spongy mesophyll layer. This loose, irregular arrangement of cells — unlike the tightly packed palisade cells above — maximises the surface area exposed to air, making gas diffusion efficient. The warm water in the experiment displaced gases already trapped in these intercellular air spaces, which is why bubbles appeared almost immediately after the leaves were submerged.

Besides stomata, older woody stems exchange gases through lenticels — small, loosely packed openings in the bark (you can see these as tiny dots on the stem of a mature bean plant or a mango tree branch). Roots exchange gases directly through the spaces between soil particles, absorbing O₂ needed for root cell respiration.

[DIAGRAM: Cross-section of a sukuma wiki leaf showing upper epidermis, palisade mesophyll, spongy mesophyll with intercellular air spaces, lower epidermis, stomata, guard cells, and vascular bundle.]

Section 2: Regulating Gas Exchange — Why Did Warm Water Produce More Bubbles?

Explain how and why the rate of gaseous exchange in plants changes. In your answer:

- Explain the role of guard cells and stomatal opening/closing in regulating gas exchange.
- Use your understanding of diffusion to explain why warm water produced more bubbles than cold water.
- Identify at least THREE environmental factors that affect the rate of gaseous exchange and explain the

The increased bubble production in warm water is explained by two linked ideas: diffusion rates and stomatal behaviour. Diffusion is the net movement of molecules from a region of high concentration to low concentration. Warm water has more kinetic energy, which means gas molecules already dissolved in or trapped within the leaf tissues move faster and escape more readily into the surrounding water — producing more visible bubbles. This is the same reason that a warm soda fizzes more vigorously than a cold one.

Guard cells regulate stomatal aperture primarily in response to light and water availability. During the day, guard cells photosynthesise, accumulate potassium ions (K⁺), and take in water by osmosis, becoming turgid. This causes stomata to open, allowing CO₂ to diffuse in for photosynthesis and O₂ to diffuse out. At night, or when the plant is water-stressed, guard cells lose K⁺ and water, become flaccid, and stomata close — reducing gas exchange and limiting water loss.

Three key environmental factors:

1. Light intensity — Higher light (e.g., midday sun on a Kiambu kale farm) triggers stomatal opening and increases photosynthesis, raising demand for CO₂ and thus the rate of gaseous exchange.

mechanism behind each. – Relate your answer to a real Kenyan farming context — for example, how a tea farmer in Kericho or a kale farmer in Kiambu might observe these effects.	2. Temperature — Warmer temperatures (up to an optimum of about 25–30°C) increase enzyme activity and diffusion rates, speeding up both photosynthesis and respiration. Extreme heat causes stomatal closure to prevent wilting, as Kericho tea farmers observe during rare heat spells. 3. Water availability — During the long dry season, plants close stomata to conserve water, drastically reducing CO₂ uptake and slowing photosynthesis. Farmers who irrigate their sukuma wiki plots maintain open stomata and higher productivity.
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Section 3: Aerobic Respiration — What Happens When Plant Cells Release Energy?	
Explain the process of aerobic respiration in plant cells. In your answer: – Write and explain the overall word equation and the balanced chemical equation for aerobic respiration. – Describe the THREE main stages of aerobic respiration (glycolysis, the Krebs cycle, and the electron transport chain), including where in the cell each stage occurs. – Explain what 'ATP' is and why it matters to a living plant cell. – Connect respiration back to the sukuma wiki phenomenon: what is actually happening inside the leaf cells that is contributing to gas production?	Every living cell — including each cell in the sukuma wiki leaves on Mwalimu Njoroge's lab bench — needs a constant supply of energy to carry out life processes such as active transport, cell division, and protein synthesis. This energy is released from glucose through aerobic respiration, summarised as: Word equation: Glucose + Oxygen → Carbon dioxide + Water + Energy Balanced equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{ATP energy}$ Aerobic respiration occurs in three stages: 1. Glycolysis (cytoplasm): One molecule of glucose (6 carbons) is split into two molecules of pyruvate (3 carbons each). This stage does not require oxygen and produces a small net gain of 2 ATP molecules. It is the only stage that can also proceed without oxygen (anaerobic conditions). 2. Krebs Cycle / Citric Acid Cycle (matrix of the mitochondrion): Pyruvate is converted to acetyl-CoA, which enters the cycle. Carbon atoms are progressively removed as CO₂ (this is the carbon dioxide that eventually diffuses out through the stomata!), and hydrogen atoms are picked up by carrier molecules (NAD and FAD). Only 2 more ATP are produced directly here, but the energy-rich hydrogen carriers are the key output. 3. Electron Transport Chain (inner mitochondrial membrane): The hydrogen carriers release their hydrogen atoms, whose electrons pass along a chain of proteins, releasing energy that is used to synthesise a large yield of ATP — up to 32–34 molecules per glucose. Oxygen acts as the final electron acceptor, combining with hydrogen to form water. ATP (adenosine triphosphate) is the universal energy currency of the cell. When ATP is broken down to ADP + phosphate, energy is released in a form the cell can use directly. A sukuma wiki plant making rapid cell growth uses enormous amounts of ATP for active transport of minerals and for building new cell walls. In the phenomenon: the CO₂ produced in the mitochondria of leaf cells diffuses into intercellular air spaces. When warm water displaces this trapped gas, we see it as bubbles. The warmer the water, the faster diffusion occurs — explaining the pattern the class observed.

Section 4: Anaerobic Respiration and the Comparison with Aerobic Respiration

Explain what happens when plant cells respire without oxygen. In your answer:

- Describe the process and products of anaerobic respiration in plant cells.
- Write the word equation for anaerobic respiration in plants.
- Compare aerobic and anaerobic respiration in a clearly organised table (consider: oxygen requirement, products, ATP yield, location, reversibility).
- Give a real-world Kenyan example of when a plant might rely on anaerobic respiration (e.g., waterlogged rice paddies near Ahero, flooded maize fields in Kisumu County after heavy rains).

When oxygen is unavailable — for example, in the waterlogged soil of the Ahero rice irrigation scheme in Kisumu County after seasonal flooding — plant root cells cannot carry out aerobic respiration. Instead, they switch to anaerobic respiration (also called fermentation in plants).

In plant cells, anaerobic respiration proceeds as follows: after glycolysis produces pyruvate, the pyruvate is converted directly to ethanol (alcohol) and carbon dioxide, regenerating NAD so that glycolysis can continue.

Word equation: Glucose → Ethanol + Carbon dioxide + (small amount of) Energy

Only 2 ATP molecules are produced per glucose — compared to up to 36 ATP in aerobic respiration — making this process far less efficient.

Comparison Table:

Feature	Aerobic Respiration	Anaerobic Respiration
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Oxygen required?	Yes	No
Final products	CO ₂ + H ₂ O	Ethanol + CO ₂ (in plants)
ATP yield	~36 ATP per glucose	2 ATP per glucose
Location	Cytoplasm + mitochondria	Cytoplasm only
Sustainability	Indefinite	Short-term only (ethanol is toxic)

Real-world Kenyan context: After the heavy long rains in Kisumu County, maize fields near Lake Victoria may become waterlogged. The roots of the maize plants are deprived of oxygen and switch to anaerobic respiration. Over time, the accumulation of ethanol in root cells becomes toxic, damaging cell membranes. This is one reason why prolonged flooding kills crops — farmers in the Kano Plains have learnt to plant on raised beds or use drainage channels to prevent root suffocation.

Section 5: Answering the Driving Question — Putting It All Together

Now bring everything together to fully answer the driving question: 'How do plants breathe, and what happens inside their cells when they release the energy stored in food?'

Your response must:

- Revisit the sukuma wiki phenomenon from Mwalimu

When Mwalimu Njoroge's students submerged their sukuma wiki leaves in warm water, they were unknowingly setting up a window into two of the most fundamental processes in all of biology: gaseous exchange and cellular respiration.

The bubbles appeared from specific spots — not the whole leaf — because gases exit the leaf almost exclusively through stomata, the microscopic pores on the lower epidermis controlled by guard cells. The gas inside those bubbles was primarily carbon dioxide (CO₂) produced by the leaf cells during cellular respiration, along with some oxygen from photosynthesis and other trapped air from the intercellular spaces. When warm water was used, more bubbles appeared because higher temperatures increase the kinetic energy of gas molecules, speeding up diffusion out of the leaf tissues — exactly as the diffusion model predicts.

Inside every sukuma wiki cell, mitochondria are continuously breaking down glucose through aerobic respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$. This three-stage process — glycolysis in the cytoplasm, the Krebs cycle in the mitochondrial matrix, and the

Njoroge's class and explain every puzzling observation the students noticed (the specific spots, the gas identity, the temperature effect). – Synthesise gaseous exchange AND respiration into one coherent explanation. – Explain the relationship between photosynthesis and respiration in terms of gas use and production (you do not need to explain photosynthesis in detail, but acknowledge its role). – Reflect on why this knowledge matters — use a specific Kenyan agricultural or environmental example to show real-world relevance.	electron transport chain on the inner mitochondrial membrane — converts the chemical energy stored in food into ATP, the usable energy currency the cell needs for growth, transport, and repair. The CO₂ released in the Krebs cycle diffuses through the spongy mesophyll air spaces and out through the stomata — this is the plant 'breathing out.' Plants also 'breathe in': stomata open during daylight to absorb CO₂ for photosynthesis, which produces the glucose that respiration will later break down. So photosynthesis and respiration are complementary cycles — one builds glucose and releases O₂, the other breaks glucose down and releases CO₂. During the day, photosynthesis dominates and the net gas exchange appears to be CO₂ in and O₂ out. At night, only respiration continues, reversing the visible gas exchange. This knowledge has urgent real-world relevance. Kenyan smallholder farmers growing sukuma wiki, maize, and beans depend on healthy leaf gas exchange for crop productivity. Understanding that stomata close under heat and drought stress helps explain why crops wilt and yields drop during the dry season in counties like Makueni or Kitui. Agricultural researchers at the Kenya Agricultural and Livestock Research Organisation (KALRO) use this knowledge to breed drought-tolerant varieties with more efficient stomatal regulation. And in waterlogged fields near Kisumu or Ahero, the switch to inefficient anaerobic respiration explains rapid crop loss after flooding. In every case, the tiny bubbles rising from Mwalimu Njoroge's sukuma wiki leaves tell a much larger story — one of energy, survival, and the invisible chemistry keeping Kenya's farms alive.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Accuracy of Scientific Content	All scientific facts are correct and precise. Equations (word and chemical) are accurate. All three stages of aerobic respiration are correctly described with locations. Stomatal mechanism, guard cell function, and diffusion are explained without errors. Anaerobic respiration products in plants are correctly stated as ethanol and CO ₂ .	Most scientific content is correct. Minor errors present (e.g., slight imprecision in ATP yield or stage location) but core concepts — respiration equation, stomatal function, aerobic vs anaerobic comparison — are accurately conveyed.	Several factual errors or misconceptions are present (e.g., confusing lactic acid fermentation with plant anaerobic respiration, incorrectly stating stomata are on the upper epidermis only, or omitting key products of respiration). Core concepts are partially understood but unreliable.
Use of Scientific Vocabulary	Consistently uses and correctly applies all key terms: stomata, guard cells, lenticels, intercellular air spaces, spongy mesophyll, glycolysis, Krebs cycle, electron transport chain, ATP, aerobic, anaerobic, diffusion, turgid, flaccid, mitochondria. Terms are embedded naturally in explanations, not just listed.	Uses most key vocabulary correctly. Occasional missing term or slight misuse (e.g., using 'breathing' without clarifying it means gaseous exchange, or omitting 'electron transport chain' by name). Vocabulary generally supports rather than hinders the explanation.	Few key terms are used correctly or consistently. Over-reliance on everyday language ('the leaf sucks in air', 'the plant breathes like us') without scientific precision. Key structures or processes are named incorrectly or not named at all.

Explanation of the Phenomenon	Every puzzling observation from the sukuma wiki experiment is explicitly and correctly explained: (1) bubbles from specific spots → stomata locations; (2) identity of gas → CO ₂ /trapped air from respiration and intercellular spaces; (3) temperature effect → increased kinetic energy and diffusion rate; (4) connection to respiration occurring in leaf cells. The phenomenon is revisited meaningfully in the final synthesis section.	Most observations are explained correctly. At least two of the four puzzling observations are linked to the correct mechanisms. The phenomenon is referenced in most sections but the final synthesis may not fully integrate all observations.	The phenomenon is mentioned but explanations are vague or partially incorrect (e.g., stating bubbles are oxygen from photosynthesis only, or that temperature 'makes the plant grow faster' without linking to diffusion or respiration rates). Fewer than two observations are correctly accounted for.
Synthesis and Connections Across Concepts	The student clearly connects gaseous exchange and respiration as related processes. The relationship between photosynthesis and respiration (gas exchange reversal between day and night) is acknowledged. Aerobic and anaerobic respiration are compared coherently. The final section weaves all unit concepts into a single, coherent answer to the driving question.	Gaseous exchange and respiration are connected in most sections. The day/night photosynthesis–respiration relationship is mentioned but may lack detail. The comparison of aerobic and anaerobic is adequate but not fully integrated into the overall narrative. The driving question is answered but with some gaps in synthesis.	Concepts are treated as separate, unrelated topics. The student explains stomata in one section and respiration in another with no explicit connection. The driving question answer in the final section does not draw on earlier sections. Little or no synthesis is evident.
Kenya Context and Real-World Application	Multiple specific Kenyan contexts are woven naturally into explanations (e.g., Kericho tea farming, Kisumu flooding and anaerobic respiration, KALRO drought-tolerant breeding, Kiambu kale farming). Examples are accurate, relevant, and used to genuinely deepen the scientific explanation rather than as decoration.	At least two Kenyan contexts are used and are broadly accurate. Examples are relevant to the biology content but may be generic (e.g., 'farmers in Kenya' without specifics) or not fully explained in terms of the underlying science.	Kenyan context is absent or only superficially mentioned (e.g., 'sukuma wiki is eaten in Kenya'). Real-world application is missing or unconnected to the scientific content of the unit.